



# Workforce Salary Analysis & Insights

SQL-Based Data Analysis Project

# Introduction

- This project analyzes workforce salary data across different roles, experience levels, and locations.
- Objective is to extract meaningful insights using SQL queries.
- Focus areas:
  - ✓ Salary trends
  - ✓ Remote work adoption
  - ✓ Role-based salary differences
  - ✓ Career growth opportunities

## Objectives

- Analyze salary distribution across roles and experience levels
- Identify high-paying countries and job roles
- Study salary growth trends over time
- Evaluate remote work adoption
- Provide career transition recommendations

# Dataset Overview

- Dataset contains workforce-related information:
- Job Title
- Experience Level
- Salary (USD)
- Company Location
- Employment Type
- Remote Ratio
- Work Year

# Methodology

- Data imported into MySQL
- Data cleaning and structuring
- SQL queries used:
  - Aggregation (AVG, COUNT)
  - Filtering (WHERE, HAVING)
  - Joins
  - CTEs
  - Window Functions



# SQL Queries >

## Task1

```
-- Task1 You need to count how many employees are working in different companies, categorized by size (S, M, L).
```

```
SELECT
    company_size,
    COUNT(employee_id) AS employee_count
FROM workforce
WHERE work_year = 2021
GROUP BY company_size;
```

## Task2

```
-- Task2 Top 3 job titles with the highest average salary for
-- part-time positions in 2023 & only include countries with more than 50 employees

SELECT
    job_title,
    AVG(salary_in_usd) AS avg_salary
FROM workforce
WHERE employment_type = 'PT'
    AND work_year = 2023
GROUP BY job_title
HAVING COUNT(*) > 50
ORDER BY avg_salary DESC
LIMIT 3;
```

## Task3

```
-- Task3  Identify countries where the average salary for mid-level
-- employees (MI) is greater than the overall average for that level in 2023:

SELECT
    company_location,
    ROUND(AVG(salary),2) AS average_salary
FROM workforce
WHERE work_year = "2023" AND experience_level = "MI"
GROUP BY company_location
HAVING AVG(salary) > (SELECT AVG(salary)
                      FROM workforce
                      WHERE work_year = "2023" AND experience_level = "MI")
ORDER BY average_salary DESC;
```

## Task4

```
-- Task4 Identify which countries pay seniorlevel (SE) employees the highest and lowest average salaries in 2023
```

```
WITH cte AS
```

```
(
```

```
SELECT
```

```
    company_location,
```

```
    AVG(salary) AS average_salary
```

```
FROM workforce
```

```
WHERE experience_level = "SE" AND work_year = "2023"
```

```
GROUP BY company_location
```

```
)
```

```
-- Country with highest average salary
```

```
SELECT * FROM cte
```

```
ORDER BY average_salary DESC
```

```
LIMIT 1;
```

```
-- Country with lowest average salary
```

```
SELECT * FROM cte
```

```
ORDER BY average_salary
```

```
LIMIT 1;
```



## Task5

```
-- Task5 Calculate the percentage increase in salaries for various job titles between two years (e.g., 2023 and 2024).

WITH cte_2023 AS
(
  SELECT
    job_title,
    ROUND(AVG(salary),2) AS avg_salary_2023
  FROM workforce
  WHERE work_year = 2023
  GROUP BY job_title
),
cte_2024 AS
(
  SELECT
    job_title,
    ROUND(AVG(salary),2) AS avg_salary_2024
  FROM workforce
  WHERE work_year = 2024
  GROUP BY job_title
)

SELECT
  cte_2023.job_title,
  avg_salary_2023,
  avg_salary_2024,
  ROUND((avg_salary_2024 - avg_salary_2023)*100.0 / avg_salary_2023,2) AS percentage_change
FROM cte_2023 JOIN cte_2024
ON cte_2023.job_title = cte_2024.job_title;
```

## Task6

```
-- Task6 Top three countries with the highest salary growth for entrylevel roles from 2020 to 2023.
```

```
WITH cte_2020 AS
(
  SELECT
    company_location,
    ROUND(AVG(salary),2) AS avg_salary_2020
  FROM workforce
  WHERE work_year = 2020 AND experience_level = "EN"
  GROUP BY company_location
),
cte_2023 AS
(
  SELECT
    company_location,
    ROUND(AVG(salary),2) AS avg_salary_2023
  FROM workforce
  WHERE work_year = 2023 AND experience_level = "EN"
  GROUP BY company_location
),

SELECT
  cte_2020.company_location,
  avg_salary_2020,
  avg_salary_2023,
  ROUND((avg_salary_2023 - avg_salary_2020)*100.0 / avg_salary_2020,2) AS percentage_change
FROM cte_2020 JOIN cte_2023
ON cte_2020.company_location = cte_2023.company_location
ORDER BY percentage_change DESC
LIMIT 3;
```

## Task7

```
-- Task7 Update remote work ratio for employees earning more than $90,000 in the US and AU

UPDATE workforce
SET remote_ratio = "100"
WHERE salary > 90000 AND employee_residence IN ("US", "AU");
```

## Task8

```
-- Task8 Update the salaries for various experience levels (SE, MI, etc.) according to predefined percentage increases
```

```
• UPDATE workforce
```

```
SET salary = salary *
```

```
  CASE
```

```
    WHEN experience_level = 'SE' THEN 1.22
```

```
    WHEN experience_level = 'MI' THEN 1.30
```

```
    WHEN experience_level = 'EN' THEN 1.35
```

```
    WHEN experience_level = 'EX' THEN 1.20
```

```
    ELSE 1
```

```
  END
```

```
WHERE work_year = 2024;
```

## Task9

```
-- Task9 Identify which year had the highest average salary for each job title.

WITH cte AS (
    SELECT
        job_title,
        work_year,
        AVG(salary_in_usd) AS avg_salary
    FROM workforce
    GROUP BY job_title, work_year
),
cte2 AS (
    SELECT
        job_title,
        work_year,
        avg_salary,
        ROW_NUMBER() OVER(PARTITION BY job_title ORDER BY avg_salary DESC) AS rnk
    FROM cte
)

FROM cte

)

SELECT *
FROM cte2
WHERE rnk = 1;
```

## Task10

```
-- Task10 Calculate the percentage of full-time and part-time employees for each job title.

SELECT
    job_title,
    ROUND(SUM(CASE WHEN employment_type = "FT" THEN 1 ELSE 0 END)*100 / COUNT(*),2) AS ft_percentage,
    ROUND(SUM(CASE WHEN employment_type = "PT" THEN 1 ELSE 0 END)*100 / COUNT(*),2) AS pt_percentage
FROM workforce
GROUP BY job_title;
```

## Task12

```
-- Task12 Top 5 countries with the most large companies

SELECT
    company_location,
    COUNT(*) AS L_companies_count
FROM workforce
WHERE company_size = 'L'
GROUP BY company_location
ORDER BY L_companies_count DESC
LIMIT 5;
```

## Task13

```
-- Task13 Calculate the percentage of fully remote employees earning more than $100,000
```

```
SELECT
```

```
    ROUND(SUM(CASE WHEN remote_ratio = 100 AND salary > 100000 THEN 1 ELSE 0 END)*100 / COUNT(employee_id),2) AS remote_emp_percentage
```

```
FROM workforce;
```



## Task14

```
-- Task14 Identify locations where entrylevel salaries surpass the market average.

SELECT
    company_location,
    ROUND(AVG(salary_in_usd),2) AS avg_en_salary
FROM workforce
WHERE experience_level = "EN"
GROUP BY company_location
HAVING avg_en_salary > (SELECT AVG(salary_in_usd) FROM workforce WHERE experience_level = "EN");
```

## Task15

-- Task15 For each job title, identify which country pays the highest average salary.

WITH avg\_cte AS

(  
 SELECT  
 job\_title,  
 company\_location,  
 ROUND(AVG(salary\_in\_usd),2) AS avg\_salary  
 FROM workforce  
 GROUP BY job\_title, company\_location  
)

,  
rnk\_cte AS

(  
 SELECT  
 \*,  
 ROW\_NUMBER() OVER(PARTITION BY job\_title ORDER BY avg\_salary DESC) AS rnk  
 FROM avg\_cte  
)

SELECT \* FROM rnk\_cte WHERE rnk = 1;

## Task16

```
-- Task16 Identify countries with consistent salary growth over the past three years.
```

```
WITH avg_cte AS
```

```
(
```

```
SELECT
```

```
    company_location,
```

```
    work_year,
```

```
    ROUND(AVG(salary_in_usd),2) AS avg_salary
```

```
FROM workforce
```

```
GROUP BY company_location, work_year
```

```
),
```

```
lag_cte AS
```

```
(
```

```
SELECT
```

```
    company_location,
```

```
    work_year,
```

```
    avg_salary,
```

```
    LAG(avg_salary) OVER(PARTITION BY company_location ORDER BY work_year) AS previous_salary
```

```
FROM avg_cte
```

```
)
```

```
SELECT
```

```
    company_location
```

```
FROM lag_cte
```

```
WHERE previous_salary IS NOT NULL
```

```
    AND avg_salary > previous_salary
```

```
GROUP BY company_location
```

```
HAVING COUNT(*) >=2;
```

## Task17

-- Task17 COMPARE THE ADOPTION OF FULLY REMOTE WORK ACROSS EXPERIENCE LEVELS BETWEEN 2021 AND 2024.

```
WITH cte_2021 AS (  
    SELECT  
        experience_level,  
        100.0 * SUM(CASE WHEN remote_ratio = 100 THEN 1 ELSE 0 END) / COUNT(*) AS remote_2021  
    FROM workforce  
    WHERE work_year = 2021  
    GROUP BY experience_level  
) ,  
cte_2024 AS (  
    SELECT  
        experience_level,  
        100.0 * SUM(CASE WHEN remote_ratio = 100 THEN 1 ELSE 0 END) / COUNT(*) AS remote_2024  
    FROM workforce  
    WHERE work_year = 2024  
    GROUP BY experience_level  
)  
  
SELECT  
    c1.experience_level,  
    ROUND(c1.remote_2021, 2) AS remote_2021,  
    ROUND(c2.remote_2024, 2) AS remote_2024  
FROM cte_2021 c1  
LEFT JOIN cte_2024 c2  
ON c1.experience_level = c2.experience_level;
```

## Task18

```
-- Task18 Average salary increase percentage by experience level and job title (2023 to 2024):
```

```
WITH cte_2023 AS
```

```
(
```

```
SELECT
```

```
    experience_level,
```

```
    job_title,
```

```
    ROUND(AVG(salary_in_usd),2) AS avg_salary_2023
```

```
FROM workforce
```

```
WHERE work_year = 2023
```

```
GROUP BY experience_level, job_title
```

```
)
```

```
,
```

```
cte_2024 AS
```

```
(
```

```
SELECT
```

```
    experience_level,
```

```
    job_title,
```

```
    ROUND(AVG(salary_in_usd),2) AS avg_salary_2024
```

```
FROM workforce
```

```
WHERE work_year = 2024
```

```
GROUP BY experience_level, job_title
```

```
)
```

```
SELECT
```

```
    cte_2023.experience_level,
```

```
    cte_2023.job_title,
```

```
    avg_salary_2023,
```

```
    avg_salary_2024,
```

```
    (avg_salary_2024 - avg_salary_2023)*100.0 / avg_salary_2023 AS percentage_change
```

```
FROM cte_2023 JOIN cte_2024
```

```
ON cte_2023.experience_level = cte_2024.experience_level
```

```
    AND cte_2023.job_title = cte_2024.job_title;
```

# Key Insights & Conclusions

- ✓ Salaries increase with experience
- ✓ Remote work is rapidly growing
- ✓ AI/ML roles offer highest salaries
- ✓ Certain countries dominate high compensation
- ✓ Entry-level roles show strong growth



**Thank You.**